

Multiple Choice

1. **Answer:** D

Options: A) exactly 10 times; B) an average of 10 times, with an rms deviation of about 0.1; C) an average of 10 times, with an rms deviation of about 1; D) an average of 10 times, with an rms deviation of about 3

Note: Correct option: D - an average of 10 times, with an rms deviation of about 3

2. **Answer:** D

Options: A) As; B) P; C) Sb; D) B

Note: Correct option: D - B

3. **Answer:** D

Options: A) independent of M ; B) proportional to $m^{(1/2)}$; C) linear in R ; D) proportional to $R^{(3/2)}$

Note: Correct option: D - proportional to $R^{(3/2)}$

4. **Answer:** D

Options: A) $(1/2)$ k T; B) kT; C) $(3/2)$ k T; D) 3 kT

Note: Correct option: D - 3 kT

5. **Answer:** A

Options: A) rate of change of the electric flux through S; B) electric flux through S; C) time integral of the magnetic flux through S; D) rate of change of the magnetic flux through S

Note: Correct option: A - rate of change of the electric flux through S

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'an average of 10 times, with an rms deviation of about 3'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'rate of change of the electric flux through S'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 24.6 eV. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 10/11 c. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key